

Interactive MObjects Tester: Algorithm and Parameters (No Spike Gate)

Reference guide for the MObjects foreground-mask backend and tuning controls, after the spike-gate comparison view was removed.

Scope: this document describes `interactive_mtobjects_parameter_tester_No_SpikeGate.py`, a variant of the MObjects tester that runs a single MObjects pass per galaxy (the spike-gate second pass and its seven tuning parameters were removed). The original `interactive_mtobjects_parameter_tester.py`, with spike gate intact, is unchanged and still available side by side. This variant keeps the same manifest, geometry, profile, and PNG review workflow as the original.

Algorithm Summary

MObjects detects astronomical sources by building a max tree of the preprocessed image and applying statistical attribute filtering to identify significant connected structures. The project README describes it as a tool for detecting sources, creating segmentation maps, and producing parameter tables; the implementation is based on the C work by Teeninga et al. and wrapped by Caroline Haigh's Python package.

1. Prepare either the residual image or original image as the detection surface.
2. Estimate or accept supplied background mean, variance, soft bias, and gain values.
3. Subtract the background, optionally smooth with a Gaussian kernel, truncate negative values, and replace NaNs for MObjects.
4. Build a max tree over the preprocessed image.
5. Run MObjects statistical filtering using `alpha`, `move_factor`, and `min_distance`.
6. Relabel the object map, restore non-finite pixels to background, and convert MObjects labels to a boolean foreground mask.
7. Apply comparison-layer filters: minimum area, maximum area, elongation, central exclusion, and mask dilation.

MObjects Parameters

Every GUI control label now states which direction increases masking aggressiveness, so the effect of a change is visible without leaving the sidebar. Defaults reflect the Optuna toy-object-recovery optimisation run of 2026-07-19 (objective = -0.304658). Toy-object optimisation runs created before

2026-07-22 are superseded, because the current optimiser injects toys only into the same deprojected, bar-aligned galaxy investigation cutout used for the normal image/profile reports.

Parameter	Default	More aggressive when...	Effect
alpha	1e-6	higher	Significance level used by the MTOjects statistical test; a higher value is a less strict threshold, so more objects are flagged.
move_factor	0.4402	lower	Controls how far object markers move upward in the tree; a lower value includes more of a large object's faint outskirts, so more pixels are masked.
min_distance	0.2367	lower	Minimum brightness separation required between objects; a lower value makes it easier for a node to pass the significance test, so more objects are flagged.
gaussian_fwhm	2.084 px	not directional	FWHM used by MTOjects preprocessing before tree construction; 0 disables smoothing. Changes sensitivity to small/faint features rather than simply increasing or decreasing masking.
soft_bias	0.0	n/a	Constant bias term used when estimating gain.
gain	-1.0	n/a	Electrons per ADU. A negative value asks MTOjects to estimate it.
bg_mean	NaN	n/a	Background mean. NaN asks MTOjects to estimate it from flat tiles.
bg_variance	-1.0	n/a	Background variance. A negative value asks MTOjects to estimate it.

Post-Detection Parameters

Parameter	Default	More aggressive when...	Effect
minarea	36 px	lower	Rejects small detected segments after MTOjects relabeling; a lower threshold keeps/masks smaller segments too.
max_area	2073 px	higher	Rejects broad segments that are more likely galaxy/bar structure than compact foreground objects; a higher cap lets larger segments be masked too.
max_elongation	15.49	higher	Rejects very elongated detections using weighted second moments; a higher cap lets more elongated shapes be masked too.
exclude_center_pixels	8 px	lower	Rejects detections close to the galaxy centre, with arcsec display conversion available; a smaller protected radius exposes more of the centre to masking.
dilation_radius	1 px	higher	Expands accepted segments before replacing/masking pixels; a larger radius grows the mask further around each segment.

Removed: Spike-Gate Parameters

The spike-gate comparison view (a second, more inclusive MTOjects pass gated to bar-major-axis profile spikes) and its seven tuning parameters (`spike_gate_move_factor`, `spike_excess_fraction`, `spike_neighbour_inner_arcsec`, `spike_neighbour_outer_arcsec`, `spike_side_offset_samples`, `spike_side_drop_fraction`, `spike_window_samples`) have been removed from both the internal code and the GUI. The figure no longer includes the third, spike-gated profile panel; the two remaining profile panels (original and MTOjects-processed) now use the freed vertical space. See `interactive_mtobjects_algorithm_and_parameters.docx` for the historical spike-gate reference.

Output Folder

PNGs from this version are written to a new folder, `interactive_mtobjects_parameter_tester_no_spike_gate` (under the machine's foreground-removal research folder), rather than the original `interactive_mtobjects_parameter_tester` folder, so outputs from before and after the spike-gate removal do not mix.

Operational Notes

- MTOBjects currently expects compiled shared libraries in the MTOBjects checkout; use `--mtobjects-root` or `MTOBJECTS_ROOT`.
- The tester treats MTOBjects as the detector and keeps comparison filters outside the library adapter.
- Mask replacement currently uses the median of finite unmasked pixels, matching the SEP tester's simple visualization-oriented cleaning.
- The MTOBjects README warns the project is a work in progress, so parameter sweeps should record library version or commit when used for comparisons.
- Parameter direction claims (`alpha`, `move_factor`, `min_distance`) are taken from the `CarolineHaigh/mtobjects` README and `mt_node_test_4.c` significance test, not re-derived empirically.

Sources

- `CarolineHaigh/mtobjects` README: source detection, segmentation maps, dependencies, and command-line parameters.
- `CarolineHaigh/mtobjects` source files: `main.py`, `preprocessing.py`, `tree_filtering.py`, `maxtree.py`, `postprocessing.py`, and `mt_node_test_4.c`.
- Local implementation: `interactive_mtobjects_parameter_tester_No_SpikeGate.py`.

Systematic documentation review - 16 August 2026

Review classification: Reviewed - interactive baseline reference remains valid

Updates made: No optimisation-result claims were found. A dated scope note was added without changing the algorithm or parameter descriptions.

Current interpretation: This document describes interactive MTOBjects without Spike Gate. The selected Toy Objects production parameters are recorded in the combined outcome document, not here.

Authoritative combined outcome: SEP and MTOjects Toy Objects Optimisation Summary and Conclusions.docx. Full-precision results: Foreground Masking Optimisation Results.xlsx.